Part One (multiple choice)

Q. One: $\int_{-\infty}^{\infty} (u(x) + \sin(x+2))\, \delta(x+2)\, dx =$

A) 1 B) 0 C) 3 D) 2 E) None

Q. Two: The mean of the process $X(t) = \cos(t+\theta)$ where $\theta \sim U(-2\pi, 2\pi)$ is

A) 1 B) $\sin(t)$ C) $\frac{1}{4\pi}\sin(t)$ D) 0 E) None

Q. Three: For $X(t) = A\cos(t)$ and $Y(t) = B\sin(t)$ where both A, B are independent random variables with equal mean. If $R_{XY}(t, t+\tau) = \cos(t)\sin(t+\tau)$ then $E(A) =$

A) 1 B) 2 C) 3 D) 4 E) None

Q. Four: The variance of a mean-ergodic process $X(t)$ with no periodic component an $R_{XX}(\tau) = Ae^{-|\tau|} + \frac{1}{1+\tau^2}$ equals 4 , then the constant A equals

A) 1 B) 2 C) 3 D) 4 E) None

ive: For the joint PDF $f(x,y) = 8b$ when $0 < x < b$, $0 < y < b$ and zero otherwise, the positive constant b equals

A) 2 B) 1/2 C) 1/3 D) 3

None

Q. Six: If $F(0) = 0$ and $P(0 < x < 1) = 1$, then $F(F(1)) =$

A) 2 B) 4 C) 3 D) 1 E) None

Q. Seven: A random variable X with pdf $f(x) = x^2$, $x \in (0, \sqrt[3]{3})$. The pdf of is

A) $3y^8$ B) $3y^6$ C) $3y^{2/3}$ D) y^8 E) None

Q. Eight: If $P(X = x) = 0.6\,\delta(x-1) + A\,\delta(x-2)$; A is a constant, then P(X

A) 0.2 B) 0.3 C) 0.4 D) 0.5 E) N

Q. Nine: For orthogonal processes $X(t) = A\cos(t)$ and $Y(t) = B\sin(t)$, the value of the product of the random variables A, B is

A) 0 B) 2 C) 3 D) 1 E) None

Q. Ten: If $Cov(X,Y)=1$, $Corr(x,y)=\rho$ and $Cov(\rho X,\rho Y)\geq 1$ then $\rho=$

A) -1 B) 1 C) 0 D) ± 1 E) None

Q. Eleven: For $x>1$, we have $2u(u(x))+3\delta(x)=$

A) 1 B) 2 C) 3 D) 0 E) None

Part Two (True / False)

Q. Twelve (✓, X) : The process $X(t)=A\cos(t)$, $A\sim U(-1,1)$ is first order stationary.

Q. Thirteen (✓, X): The joint CDF $F_{XY}(x,y)=e^{-y}\cos(x)$ gives $f(x,y)=e^{-y}\cos(x)$

Q. Fourteen (✓, X): If $\bar{X}=1$, $Var(X)=2$ and $Y=X+1$, then X,Y are orthogonal r.v's.

Q. Fifteen (✓, X) : If $Cov(-X,Y)=3Corr(-X,Y)$, then $\sigma_X\sigma_Y=3$

Q. Sixteen (✓, X) : For the joint PDF $f(x,y)=0.5$ when $0<x<1$, $0<y<2$ and zero otherwise, we have $f_X(x)=0.5$

Q. Seventeen (✓, X): For two independent r.vs X,Y we have $E(XY)=2E(X)$, then $E(Y)=0.5$

Q. Eighteen (✓, X): If $Var(E(X)X)=0$, then either $E(X)=0$ or $Var(X)=0$

Q. Nineteen (✓, X): $X\sim B(n,0.5)$ and $Y\sim P(2.3)$ cannot have the same means

Q. Twenty (✓, X): Two independent random variables X,Y with marginal pdf's $f_X(x)=2$ when $x\in(0,0.5)$ and $f_Y(y)=1$ when $y\in(0,1)$. Then the joint pdf is $f(x,y)=1$ when x and y belong to their intervals and zero otherwise

1) $\int_{-\infty}^{\infty} u(x)\,\delta(x-(-2))\,dx + \int_{-\infty}^{\infty} \sin(x+2)\,\delta(x-(-2))\,dx$

$= u(-2) + \sin(-2+2) = 0$

Remember: $u(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$

B

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2) $\sin(t+\theta) * \frac{1}{4\pi}\Big|_{-2\pi}^{2\pi} = \frac{1}{4\pi}(0) = 0$ (D)

3) $E(A) = E(B)$

$R_{xy}(t, t+\tau) = \cos(t)\cdot\sin(t+\tau)$

$E(x(t))\cdot E(y(t)) = \cos(t)\cdot\sin(t+\tau)$

$E(A\cos(t))\cdot E(B\cdot\sin(t+\tau)) = \cos(t)\cdot\sin(t+\tau)$

$E(A)\cdot E(B) = 1$

$E(A) = 1$ (A)

4) $R_{xx}(0) = E(x^2(t)) =$

$= A + 1 = 4 \Rightarrow$ [illegible] $A = 3$ (C)

5) $\int_0^b \int_0^b 8b \cdot dy \cdot dx \rightarrow \int_0^b 8b^2 \cdot dx$ (B)

$\int_0^b dy \cdot dx = 1$

$8b^3 = 1 \Rightarrow b = \frac{1}{2}$

6) $F(1) - F(0) = 1$

$F(1) = 1$

$F(F(1)) = F(1) = 1$ (D)

7) $f(x) = x^2$ P.d.f $\quad y = \sqrt[3]{x} \Rightarrow x = y^3 \Rightarrow \frac{dx}{dy} = 3y^2$ Find $f_y(y)$?

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$$f_y(y) = f_x(T^{-1}(y)) \cdot \left| \frac{d(T^{-1})}{dy} \right|$$

$$= y^6 \cdot 3y^2 \Rightarrow \boxed{3y^8} \ (A)$$

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8) $P(X=x) = 0.6\,\delta(x-1) + A\,\delta(x-2)$, $\quad P(X=2) = ??$

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9) $R_{xy} = 0 \Rightarrow E(x(t) \cdot y(t)) = 0$

$E(A \cos(t) \cdot B \sin(t+\tau)) = 0$

$\cos(t) \cdot \sin(t+\tau) \cdot E(AB) = 0$ $\quad E(A \cdot B) = 0$

$A \cdot B = 0$ (A)

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10) $Cov(\rho X, \rho y) \geq 1 \Rightarrow \rho^2 \cdot Cov(x,y) \geq 1$

$\rho^2 \geq 1 \Rightarrow \rho \geq |1|$

(D) $\rho = \pm 1$

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11) $2u(1+3) = 2u(4) = 2 \cdot 1 = \boxed{2}$ (B)

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(12) ✓ , (13) محذوف , (14) ✗ , (15) ✓ , (16) ✗

(17) ✗ , (18) ✓ , (19) ✓ , (20) ✗